

simulator that supports scheduling, deadlocks and advanced memory management. Zenmap is a network-mapping tool that is useful for security auditing and network exploration. Onama is also equipped with Parallel Python, a Python module that can be deployed across systems with multiple processors or cores.

In addition, Onama consists of numerous parallel and CUDA-based software, which unleash the true power of HPC. Some of the basic parallel programs include parallel matrix—matrix multiplication with different algorithms, sorting programs, etc.

Onama and open source

The Onama applications included essentially consist of open source tools and libraries. Until a few years ago, major corporations considered open source developers to be outlaws, and their creations were rebuffed without any enthusiasm. However, this scenario is rapidly changing thanks to Linux, OpenOffice.org, and other similar initiatives, which have become increasingly popular not only among freelance developers, but also among the general public. The biggest reason for this is the approach of open source developers and community members towards their products. The main objective of open source development is to create effective and robust software that can be used by people all over the world. On the other hand, software companies concentrate mostly on maximising their profits and user base.

Open source applications have a

number of advantages over traditional proprietary software. The source code is freely available, and can be customised to suit the institution's needs. The associated monetary costs are also negligible when compared to the industry standards, because of the open development. However, open source technologies do suffer from two major drawbacks. Compared to industry standards, the development is very volatile, and there is rarely any professional support available. More often than not, users are left at the mercy of the lead developers, as the community's interest dies down. However, almost all software included in the suite has been under active development for a long time. As the institutes and students become a part of the community, they can help streamline and further this development. Thus, Onama offers a number of advantages for interested institutes, engineering students, as well as open source enthusiasts.

A number of key issues and trends in HPC will impact the delivery of breakthrough science and engineering in the future. For instance, modelling and simulation are playing a greater role in all scientific disciplines—be it understanding the molecular processes in cells, or designing of next-generation batteries for hybrid vehicles. For many sectors, computing is not only the fastest and most cost-effective tool for discovery, it is the only one. And Onama will definitely help our budding engineers prepare for the future. 

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A technophile with strong writing skills, the author revels in exploring new technologies and solutions. An avid reader on open source solutions, Sanket has contributed code as well as documentation to open source projects as well as magazines. For him, open source is not just a pragmatic methodology; it is a philosophy and a way of life.

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